Max Ellis *BDes (Hons) MSc AFHEA*

PhD Candidate in Psychoacoustics

Psychoacoustic Assessment and Machine Learning-Based Modelling of Novel Noise Source Perception in Complex Soundscapes

ellisacoustics.github.io · linkedin.com/in/psychoacoustic · github.com/ellisacoustics · orcid.org/0009-0007-5909-854X

Net Zero technology will bring *sounds* we've never lived with into our towns and cities: **electric cars**, **delivery drones**, **air taxis** and **heat pumps**. I research how these noise sources affect us, *before* they arrive.

Noise is a "silent killer", second only to air pollution in its burden on health in Europe[1]. The next wave of technology will introduce noise that loudness alone cannot accurately judge[2]: the high-pitched whine of a rotor, the tonal hum of a compressor, rumbles of large aircraft[3, 4]. My PhD research, sponsored by the Department for Environment, Food and Rural Affairs (Defra), seeks to accurately simulate these future soundscapes to gather response data from listening panels, for machine learning models to predict who will be annoyed, to what extent, and why. These results give policymakers & regulators evidence to set standards on, and engineers targets for quieter machines, before they roll out en masse.

The Papers

2026

Comparing a General Sound Quality Metric for Aircraft Noise Using a Pre-Trained Neural Network Model to Traditional Psychoacoustics

T. Lobato, T. Kamper-Schley, R. Sottek, M. C. Green, **M. W. Ellis**, A. J. Torija

Quiet Drones 2026, Delft. Human Response to UAS and UAM Noise. With HEAD Acoustics GmbH.

PDF

2025

Comparison of Deep Learning and Psychoacoustic Models to Predict UAV Noise Impact in Soundscapes

M. W. Ellis, M. C. Green, M. J. B. Lotinga, A. J. Torija

Forum Acusticum EuroNoise 2025, 11th Convention of the European Acoustics Association, Málaga, pp. 2121–2127.

doi:10.61782/fa.2025.0235 · PDF

Industry Experience

Aug 2026 – Oct 2026

Specialist Consultant (PhD Placement)

Noise Consultants Ltd. (part of Logika Group)

- Validated open-source noise measurements against modelled Defra noise-mapping data for infrastructure assessment.
- Ran a field campaign on an airport expansion project comparing physical and perceptual dominance of aircraft noise.

Academic Experience

2026 – Present

Associate Lecturer in Computer Science

St Mary's University, Twickenham

- Teaching, marking and moderation across the online MSc Computer Science suite.
- Module reviewer and author.

2026 – Present

Teaching Assistant in Applied Acoustics

University of Salford

- Lab demonstrator for Environmental Noise Measurement and Modelling (BEng/MSc).

Education

2025 – 2028

PhD Acoustics and Audio Engineering

University of Salford Acoustics Innovation Institute

Year 2

2023 – 2024

MSc Computer Science with Artificial Intelligence

Abertay University

Distinction

2021 – 2023

BDes (Hons) Sound for Moving Image

University of Glasgow

2nd Class Honours, Upper-Division (2:1)

2019 – 2021

DipHE Music Technology

University of the West of Scotland

Distinction

Awarded BSc Level 8 Court Medal

2019

BTEC Level 3 Digital Synthesis and Sampling

Fife College

Pass

2018 – 2019

SCQF Level 6 Sound Production

Glasgow Clyde College Anniesland

Pass

2018 – 2019

AdvH Music

Eastwood High School

Grade: A

Awarded Eastwood Prize for Higher Music Technology

Working Groups and Committees

Institute of Acoustics Scottish Branch Committee — Member; Early Careers Group — Member

NASA Unmanned Aircraft Systems Noise Working Group (UNWG) — Member

Noise Network Plus A.I. and Digital Working Group — Member

Software Proficiencies

Max/MSP Ableton ArtemiS SUITE R Python QGIS LaTeX (Overleaf) REAPER MATLAB TensorFlow PyTorch Feature Manipulation Engine (FME)

Certifications

Advanced Prompt Engineering
Vanderbilt University
Show credential ↗

Advanced R Programming
Johns Hopkins University
Show credential ↗

AI for Noise Pollution Control
LSIB
Show credential ↗

Applications of AI in Acoustics
European Acoustics Association (EAA)
Show credential ↗

Auralization
Autumn School Series in Acoustics (ASSA)

Digital Signal Processing
EPFL
Show credential ↗

Ethics of Artificial Intelligence
Politecnico di Milano
Show credential ↗

GPU Programming
Johns Hopkins University
Show credential ↗

Mathematics for Machine Learning
Imperial College London
Show credential ↗

Responsible Research and Innovation
ORBITRRI
Show credential ↗

Transformer Models and BERT Model
Google
Show credential ↗

Trustworthy Generative AI
Vanderbilt University
Show credential ↗

Research Collaborations

Defra — PhD Sponsor

HEAD Acoustics GmbH — Conference Paper

References

1 WHO Regional Office for Europe (2011). New evidence from WHO on health effects of traffic-related noise in Europe. News release, 30 March 2011.

2 House of Lords Science and Technology Committee (2023). The effects of artificial light and noise on human health. Oral evidence, 28 February 2023.

3 Civil Aviation Authority (2018). Aircraft Noise and Annoyance: Recent findings. CAP 1588.

4 Department for Transport and Civil Aviation Authority (2026). Aviation Night Noise Effects (ANNE) study. 10 June 2026.

5 Davies, P. (2007). Perception-based engineering: integrating human responses into product and system design. *The Bridge* 37(3), 18–24. National Academy of Engineering.

6 Torija, A. J., Krishnamurthy, S. and Merino-Martinez, R. (2026). Editorial: perception-driven acoustic engineering. *Frontiers in Acoustics* 4, 1802446. doi:10.3389/facou.2026.1802446

7 University of Salford. Sound quality metrics. An introduction to sound quality testing, Acoustics Research Centre.